

HR & SALES PERFORMANCE COMPANY OVERVIEW ANALYSIS

Complete End-to-End Documentation: Data Cleansing, Advanced SQL
Business Intelligence, Relational Modeling & Power BI Dashboard Architecture

An exhaustive, step-by-step project manual evaluating corporate workforce capacity, commercial transactions, multi-dimensional sales returns, employee salary ROI, and executive business reporting.

Microsoft Excel

PostgreSQL / SQL

Microsoft Power BI

DAX Modeling

Relational Schema Design

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COVERED ASSETS & DELIVERABLES

Company Overview Analysis.sql

Employee Overview Dshboard.pbix (v1 & v2)

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Document Scope & Purpose

This documentation serves as an exhaustive technical manual and executive presentation for the analytical pipeline built upon Company Overview Analysis.sql, Employee Overview Dshboard.pbix, and their source transactional tables. It outlines data contracts, relational joins, complex SQL aggregations, DAX formulas, and managerial insights.

1. Project Overview & Executive Summary

The **HR & Sales Performance Company Overview Analysis** project represents an enterprise-grade analytics solution designed to bridge the operational gap between talent management and commercial performance. Traditional corporate reporting frequently silos human resource metrics (such as compensation, tenure, appraisal ratings, and departmental headcounts) from commercial transactional performance (including billing volume, realized net profit, discount markdowns, and client acquisition).

By unifying these operational domains within a normalized relational schema, this project delivers comprehensive visibility into corporate productivity, departmental efficiency, territorial expansion patterns, and employee-level Return on Investment (Salary ROI).

TOTAL HEADCOUNT	GROSS REVENUE	REALIZED PROFIT	AVG BASE SALARY	AVG APPRAISAL
100	₹5.17M	₹790.02K	₹79,883	3.99 / 5.0
5 Functional Units	1,000 Transactions	15.27% Net Margin	₹40.2K – ₹119.0K Range	Annual Review Index

2. Business Objectives & Analytical Framework

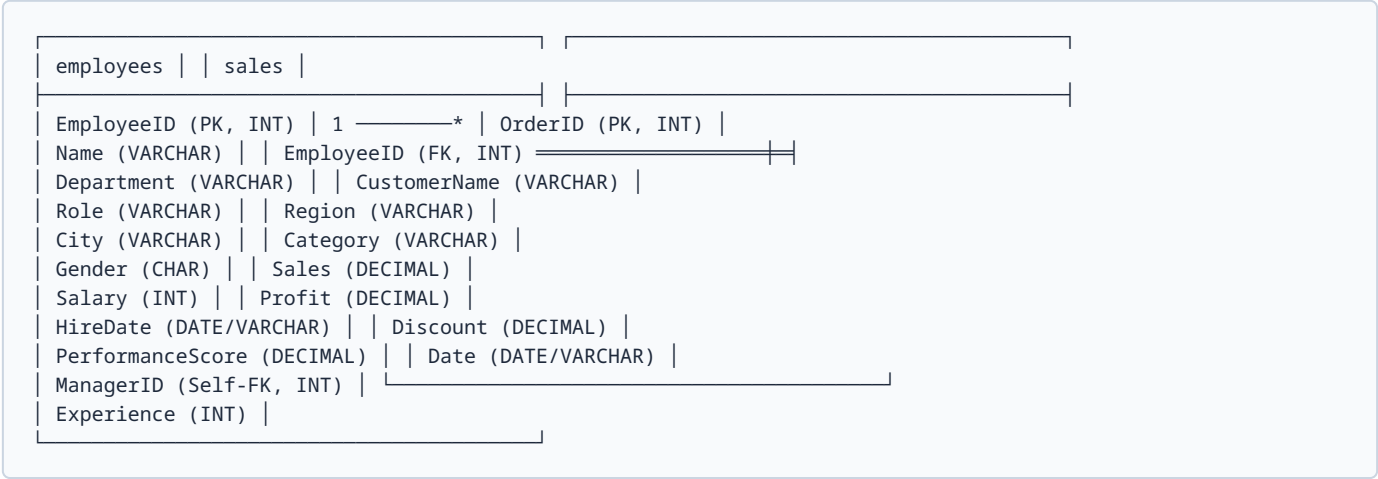
The strategic and technical objectives established for this enterprise analytics implementation comprise:

- **Workforce Capacity & Compensation Audit:** Evaluate staffing capacity, organizational hierarchy, salary distributions, and compensation parity across functional departments and regional corporate offices.
- **Commercial Transaction Profiling:** Measure gross transaction volumes and bottom-line profit contributions across geographic sales zones (North, South, East, West) and product offering categories.
- **Cross-Functional Attribution Modeling:** Establish transactional linkages between individual personnel and closed sales order revenues to calculate commercial Return on Compensation (Salary ROI).
- **Product Category Margin & Discount Auditing:** Identify high-margin versus margin-dilutive product lines, evaluating discount erosion across transactional cohorts.
- **Interactive BI Cockpit Deployment:** Translate complex analytical findings into an intuitive, multi-dimensional Microsoft Power BI dashboard featuring dynamic slicing, cross-filtering, and responsive KPI strips.

3. Dataset Architecture & Data Dictionary

The analytical data repository consists of two primary relational entities: the workforce master table (employees) and the operational commercial order table (sales).

3.1 Relational Entity-Relationship Diagram (ERD)



3.2 Data Dictionary: `employees` Table (100 Records)

Column Name	Data Type	Constraints	Sample	Business Description
EmployeeID	Integer	Primary Key, Non-null	1, 35	Unique employee corporate identifier.
Name	VARCHAR(100)	Non-null	Shreya_1	Employee legal full name / identifier.
Department	VARCHAR(50)	Non-null	Sales, IT, HR	Business unit (Marketing, IT, Sales, HR, Finance).
Role	VARCHAR(100)	Non-null	Sales Executive	Operational title designation (15 distinct roles).
City	VARCHAR(50)	Non-null	Bangalore, Noida	Base corporate office (6 major metropolitan hubs).
Gender	CHAR(1)	Categorical	M, F	Gender classification (53 Female, 47 Male).
Salary	Integer	Currency (INR)	55800, 116800	Annual base compensation (₹40,200 to ₹119,000).
HireDate	Date / Text	Non-null	30-09-2017	Formal company onboarding date (DD-MM-YYYY).
PerformanceScore	Decimal(2,1)	Range: 1.0 - 5.0	4.9, 3.3	Annual appraisal evaluation score.
ManagerID	Integer	Foreign Key	4, 7	EmployeeID of direct line manager.
Experience	Integer	Tenure (Years)	3, 12	Total career domain tenure in years.

3.3 Data Dictionary: `sales` Table (1,000 Records)

Column Name	Data Type	Constraints	Sample	Business Description
OrderID	Integer	Primary Key, Non-null	1001, 1002	Unique sales transaction line identifier.
EmployeeID	Integer	Foreign Key, Non-null	35, 44	Attributed personnel closing/servicing transaction.
CustomerName	VARCHAR(150)	Non-null	Neha Kapoor	Purchasing corporate or individual client.
Region	VARCHAR(20)	Non-null	West, East	Operating sales territory (East, North, South, West).
Category	VARCHAR(50)	Non-null	Software	Product offering segment (5 distinct categories).
Sales	Decimal(10,2)	Monetary (INR)	4637.96	Gross order billing value (₹515.02 to ₹9,999.78).
Profit	Decimal(10,2)	Monetary (INR)	909.99	Net bottom-line profit realized on transaction.
Discount	Decimal(3,2)	Percentage Rate	0.05, 0.16	Contractual discount concession (0.00 to 0.20).
Date	Date / Text	Non-null	10-10-2024	Fulfillment date timestamp (DD-MM-YYYY).

4. Technology Stack & Implementation Tools

Tool / Technology	Functional Project Role	Core Capabilities Demonstrated
Microsoft Excel	Data ingestion, hygiene auditing, pivot profiling, baseline KPI validation.	Column typing, missing record reconciliation, pivot cross-tabulation, formula validation.
PostgreSQL / SQL	Relational data modeling, cohort grouping, joins, analytical queries.	Relational Joins, Window Functions ('RANK() OVER'), CTEs, `CASE` classification, multi-tier aggregation.
Microsoft Power BI	Semantic data modeling, DAX measures, interactive executive cockpit.	Star-schema relationships, multi-card KPI visuals, clustered bar/column visuals, dynamic cross-filters.

5. End-to-End Analytics Workflow

The analytical implementation adheres to an enterprise data lifecycle methodology designed to guarantee reproducibility, structural fidelity, and auditability across all reporting tiers:

Step 1

Raw Data Ingestion & Cleansing (Excel): Ingest raw CSV sources, audit data types, standardize date serialization (`DD-MM-YYYY`), and verify key referential integrity.

Step 2

Exploratory Data Analysis (Excel Pivot Tables): Build summary cross-tabs to establish baseline benchmarks: ₹5.17M gross revenue, ₹790.02K net profit, and 100 staff headcount.

Step 3

Relational Querying & Business Intelligence (SQL): Deploy relational schema inside PostgreSQL, implementing 5 modules of structured queries in `Company Overview Analysis.sql`.

Step 4

Data Modeling & DAX Measures (Power BI): Establish active 1-to-many relationship (`employees[EmployeeID]` to `sales[EmployeeID]`), building calculated measures for executive KPIs.

Step 5

Visual Deployment & Insights Synthesis: Build responsive visual canvas in Power BI with multi-attribute slicers, translating trends into actionable business strategies.

6. Data Cleaning & Excel Exploratory Analysis

The exploratory phase ensured that both raw CSV datasets complied with strict data governance and validation standards prior to relational database migration:

- **Structural Integrity Check:** Verified that `Employees.csv` contained exactly 100 discrete rows with zero null entries in primary key (`EmployeeID`) and compensation fields.
- **Referential Consistency Audit:** Confirmed that every transactional line item in `Sales.csv` mapped to a valid `EmployeeID` between 1 and 100, preventing orphaned records.
- **Range & Anomaly Detection:** Evaluated minimum and maximum values across numerical fields. Salaries ranged from ₹40,200 to ₹119,000; transaction order values ranged from ₹515.02 to ₹9,999.78; and discounts were bound between 0.00 and 0.20.
- **Baseline Departmental Aggregation:** Constructed preliminary Pivot Tables in Excel to verify departmental headcounts: Marketing (23), IT (22), Sales (21), HR (17), and Finance (17).

7. SQL Business Intelligence & Analytical Queries

The production script `Company Overview Analysis.sql` contains 26 structured SQL queries organized into five strategic business intelligence modules.

7.1 Module 1: Workforce & HR Analytics

This module audits departmental staffing capacity, compensation distributions, performance appraisals, and geographic talent allocation.

Query 1.1: Headcount and Average Salary by Department

```
SELECT
    Department,
    COUNT(EmployeeID) AS total_employees,
    ROUND(AVG(Salary), 2) AS avg_salary
FROM employees
GROUP BY Department
ORDER BY avg_salary DESC;
```

Department	Headcount	Avg Salary (₹)	Median Salary (₹)	Comp Variance vs Org Avg
Finance	17	84,917.65	89,000.00	+6.30%
IT	22	84,159.09	85,950.00	+5.35%
Sales	21	83,395.24	78,000.00	+4.40%
Marketing	23	76,643.48	78,700.00	-4.06%
HR	17	69,358.82	62,300.00	-13.17%

Query 1.2: Average Performance Score by Department

```
SELECT
    Department,
    ROUND(AVG(PerformanceScore), 2) AS avg_performance_score
FROM employees
GROUP BY Department
ORDER BY avg_performance_score DESC;
```

Output Findings: Marketing demonstrates the highest average employee appraisal rating at **4.17**, followed by Finance at **4.07**, HR at **3.92**, IT at **3.91**, and Sales at **3.90**.

Query 1.3: Geographic Talent Allocation Across Corporate Hubs

```
SELECT
    City,
    COUNT(EmployeeID) AS total_employees,
    ROUND(AVG(Salary), 2) AS avg_salary
FROM employees
GROUP BY City
ORDER BY total_employees DESC;
```

Corporate City Hub	Employee Count	Headcount Share	Average Salary (₹)
Bangalore	23	23.0%	85,160.87
Gurgaon	20	20.0%	83,985.00
Mumbai	18	18.0%	71,100.00
Noida	17	17.0%	81,976.47
Delhi	13	13.0%	72,061.54
Pune	9	9.0%	82,188.89

7.2 Module 2: Regional Sales & Margin Performance

Evaluates commercial traction, operational efficiency, and pricing concessions across the four geographic territories.

Query 2.1: Total Sales, Profit, and Margin by Region

```
SELECT
  Region,
  ROUND(SUM(Sales), 2) AS total_sales,
  ROUND(SUM(Profit), 2) AS total_profit,
  ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS profit_margin_pct
FROM sales
GROUP BY Region
ORDER BY total_sales DESC;
```

Geographic Region	Orders	Total Sales (₹)	Total Profit (₹)	Profit Margin %	Avg Discount %
South	258	1,372,223.31	203,341.20	14.82%	10.33%
North	248	1,332,194.53	205,570.19	15.43%	9.98%
East	248	1,271,727.34	191,182.54	15.03%	9.67%
West	246	1,197,683.40	189,922.19	15.86%	9.73%
Total / Org Average	1,000	₹5,173,828.58	₹790,016.12	15.27%	9.93%

Query 2.2: Identifying Outperforming Regions Above Regional Average

```
SELECT
  Region,
  ROUND(SUM(Sales), 2) AS total_sales
FROM sales
GROUP BY Region
HAVING SUM(Sales) > (
  SELECT AVG(region_total)
  FROM (SELECT SUM(Sales) AS region_total FROM sales GROUP BY Region) AS sub
)
ORDER BY total_sales DESC;
```

Business Insight: The benchmark average regional revenue is ₹1,293,457.15. Both the **South** (₹1.37M) and **North** (₹1.33M) regions exceed this threshold, while the West region exhibits the company's highest net margin efficiency at **15.86%**.

7.3 Module 3: Product Category Intelligence

Examines commercial billing volume, realized profit, and discount sensitivity across product lines.

Query 3.1: Sales, Profit, and Operating Margin by Category

```
SELECT
  Category,
  ROUND(SUM(Sales), 2) AS total_sales,
  ROUND(SUM(Profit), 2) AS total_profit,
  ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS profit_margin_pct
FROM sales
GROUP BY Category
ORDER BY total_sales DESC;
```

Product Category	Orders	Total Sales (₹)	Total Profit (₹)	Profit Margin %	Avg Discount %
Software	236	1,211,686.62	185,585.84	15.32%	10.33%
Services	199	1,071,611.56	162,382.42	15.15%	9.47%
Office Supplies	204	1,053,570.00	161,255.33	15.31%	10.13%
Electronics	189	987,743.70	151,363.23	15.32%	9.72%
Furniture	172	849,216.70	129,429.30	15.24%	9.92%

Query 3.2: Product Category Profitability Classification

```
SELECT
  Category,
  ROUND(SUM(Profit), 2) AS total_profit,
  CASE
    WHEN SUM(Profit) >= 180000 THEN 'High Profit'
    WHEN SUM(Profit) BETWEEN 150000 AND 179999 THEN 'Moderate Profit'
    ELSE 'Low Profit'
  END AS profit_category
FROM sales
GROUP BY Category
ORDER BY total_profit DESC;
```

Outcome: *Software* emerges as the sole **High Profit** category (> ₹180K profit), followed by *Services*, *Office Supplies*, and *Electronics* in the **Moderate Profit** tier, while *Furniture* falls into the **Low Profit** tier due to lower order volume.

7.4 Module 4: Cross-Functional Employee ROI & Attribution

This relational join connects sales and employees via EmployeeID to evaluate commercial productivity by department, top individual contributors, and salary Return on Investment (ROI).

Query 4.1: Sales and Profit Attributed to Employee Departments

```
SELECT
  e.Department,
  COUNT(s.OrderID) AS total_orders_handled,
  ROUND(SUM(s.Sales), 2) AS total_sales,
  ROUND(SUM(s.Profit), 2) AS total_profit
FROM sales s
JOIN employees e ON s.EmployeeID = e.EmployeeID
GROUP BY e.Department
ORDER BY total_sales DESC;
```

Employee Department	Orders Closed	Total Sales (₹)	Total Profit (₹)	Revenue Share	Profit Margin %
Marketing	235	1,221,029.71	187,657.45	23.60%	15.37%
IT	233	1,192,402.89	176,930.16	23.05%	14.84%
Sales	201	1,044,388.73	162,831.95	20.19%	15.59%
HR	171	882,029.50	135,957.87	17.05%	15.41%
Finance	160	833,977.75	126,638.69	16.12%	15.18%

Query 4.2: Top 10 Employees by Sales & Salary ROI

```
SELECT
  e.EmployeeID, e.Name, e.Department, e.Role,
  COUNT(s.OrderID) AS orders_closed,
  ROUND(SUM(s.Sales), 2) AS total_sales,
  ROUND(SUM(s.Profit), 2) AS total_profit,
  e.Salary,
  ROUND(SUM(s.Sales) / e.Salary, 2) AS sales_to_salary_ratio
FROM sales s
JOIN employees e ON s.EmployeeID = e.EmployeeID
GROUP BY e.EmployeeID, e.Name, e.Department, e.Role, e.Salary
ORDER BY total_sales DESC
LIMIT 10;
```

ID	Employee Name	Department	Role Designation	Orders	Sales (₹)	Profit (₹)	Salary (₹)	ROI Ratio
59	Manav_59	IT	Data Analyst	18	111,363.93	17,177.25	73,800	1.51x
95	Isha_95	Marketing	Marketing Manager	14	86,546.22	12,617.39	112,100	0.77x
41	Sameer_41	Marketing	Content Strategist	13	83,551.60	13,651.23	79,500	1.05x
82	Amit_82	HR	HR Executive	14	82,677.86	14,136.71	40,900	2.02x
65	Shruti_65	Marketing	Content Strategist	14	78,769.32	13,380.64	82,900	0.95x
18	Aditya_18	Marketing	Content Strategist	12	78,323.33	12,915.86	50,600	1.55x
81	Sonia_81	IT	Software Engineer	15	75,305.01	11,154.48	66,700	1.13x
38	Simran_38	Finance	Financial Analyst	10	74,922.88	11,079.99	89,000	0.84x
90	Neha_90	IT	Data Analyst	12	73,556.54	9,987.57	105,700	0.70x
36	Aditya_36	HR	Recruiter	15	73,139.27	10,703.06	48,600	1.50x

7.5 Module 5: Customer Accounts & Multi-Year Time Trends

Profiles key corporate account revenue concentrations and historical performance trajectories from 2020 through 2024.

Query 5.1: Top 10 Customer Accounts by Revenue

```
SELECT
  CustomerName,
  COUNT(OrderByID) AS total_purchases,
  ROUND(SUM(Sales), 2) AS total_spent,
  ROUND(SUM(Profit), 2) AS total_profit_generated
FROM sales
GROUP BY CustomerName
ORDER BY total_spent DESC
LIMIT 10;
```

Customer Account Name	Transactions	Total Spend (₹)	Total Profit (₹)	Account Margin %
Sneha Patel	108	566,590.42	87,018.84	15.36%
Ananya Gupta	109	548,537.06	86,108.15	15.70%
Arjun Mehta	105	541,859.42	82,277.96	15.18%
Rahul Sharma	100	539,254.85	80,241.84	14.88%
Amit Bansal	108	527,611.81	78,505.01	14.88%
Simran Kaur	94	514,492.03	72,669.71	14.12%
Neha Kapoor	98	509,441.43	73,905.30	14.51%
Rohit Verma	99	492,821.54	77,663.15	15.76%
Karan Jain	93	487,205.51	82,328.89	16.90%
Priya Singh	86	446,014.51	69,297.27	15.54%

Query 5.2: Multi-Year Longitudinal Performance (2020 – 2024)

```
SELECT
  SUBSTR(Date, 7, 4) AS order_year,
  COUNT(OrderByID) AS total_orders,
  ROUND(SUM(Sales), 2) AS total_sales,
  ROUND(SUM(Profit), 2) AS total_profit
FROM sales
GROUP BY SUBSTR(Date, 7, 4)
ORDER BY order_year ASC;
```

Order Year	Orders Closed	Gross Revenue (₹)	Net Profit (₹)	Profit Margin %	YoY Sales Growth
2020	194	976,090.90	155,065.05	15.89%	-
2021	197	998,731.79	146,319.00	14.65%	+2.32%
2022	199	1,044,814.83	157,774.89	15.10%	+4.61%
2023	210	1,143,544.39	179,713.60	15.72%	+9.45%
2024	200	1,010,646.67	151,143.58	14.96%	-11.62%

8. Power BI Dashboard Development & DAX Specifications

The interactive report file Employee Overview Dshboard.pbix consolidates the cleansed data models into a production-ready executive dashboard built using Microsoft Fluent 2 design language.

8.1 Core DAX Business Measures

```
// 1. Total Active Headcount
Total Employees = COUNTROWS('Employees')

// 2. Aggregate Gross Commercial Sales
Total Sales = SUM('Sales'[Sales])

// 3. Aggregate Realized Net Profit
Total Profit = SUM('Sales'[Profit])

// 4. Mean Annual Base Compensation
AVG Salary = AVERAGE('Employees'[Salary])

// 5. Mean Workforce Performance Score
Avg Performance = AVERAGE('Employees'[PerformanceScore])

// 6. Net Realized Margin Ratio
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

8.2 Dashboard Visual Architecture & Container Layout (`Page 1`)

Dashboard Component	Visual Type	Bound Data Fields	Executive Reporting Purpose
Multi-Metric KPI Banner	cardVisual	`Total Sales`, `Total Profit`, `Total Employees`, `AVG Salary`, `Avg Performance`	Provides instant c-suite visibility into top-level financial health and workforce metrics.
AVG Salary By Department	clusteredBarChart	X: `[AVG Salary]` Y: `Department`	Audits departmental compensation parity (Finance & IT lead; HR trails).
Total Sales By Department	donutChart	Legend: `Department` Values: `[Total Sales]`	Highlights functional revenue share (Marketing leads at 23.60%).
Total Sales BY Region	pieChart	Legend: `Region` Values: `[Total Sales]`	Demonstrates territorial market share distribution across operating zones.
Total Profit BY Region	clusteredColumnChart	X: `Region` Y: `[Total Profit]`	Compares territorial earnings strength (North leads at ₹205.57K).
Monthly Sales Trend	lineChart	X: `Date.Month` Y: `[Total Sales]`	Tracks sales seasonality and cyclic peaks across calendar months.
Interactive Filter Panel	advancedSlicerVisual	`Year`, `Region`, `Category`, `City`, `Department`	Allows interactive slicing across temporal, geographic, and organizational dimensions.

Interactive Cross-Filtering Design

Cross-filtering is configured to respect the 1-to-many relationship: selecting an organizational dimension (e.g. *Department = 'IT'*) immediately recalculates the *Sales by Region*, *Monthly Trend*, and *Profit* visuals, isolating the transactions generated by IT department personnel.

9. Key Business Insights & Findings

Synthesis of the SQL analytical findings and Power BI visual dashboards reveals five critical strategic insights:

1. Financial & Commercial Yield Stability

Across 1,000 fulfilled transactions between 2020 and 2024, the enterprise generated **₹5,173,828.58** in gross commercial sales and realized **₹790,016.12** in net profit, maintaining an exceptionally steady operating margin of **15.27%**. Annual revenues peaked in 2023 at ₹1.14M (+9.45% YoY growth).

2. Territorial Commercial Asymmetry

Geographical performance reveals distinct strengths across regions. The **South Region** dominates total billing volume (₹1.37M, 258 orders), while the **North Region** generates the highest absolute profitability (₹205.57K). Notably, the **West Region** operates with the highest net margin efficiency at **15.86%**, supported by disciplined discounting (9.73%).

3. Software & Services Revenue Dominance

Software is the company's flagship product category, capturing ₹1.21M in sales (23.4% revenue share) and ₹185.59K in profit. Combined with **Services** (₹1.07M), intangible solutions generate 44.1% of total enterprise billing. **Furniture** lagged behind with ₹849.22K in sales due to lower order velocity (172 orders).

4. Departmental Compensation vs. Performance Disparities

Departmental compensation benchmarks reveal significant variance. **Finance** (₹84,918) and **IT** (₹84,159) command the highest base compensation, whereas **HR** trails substantially at ₹69,359 (-13.2% vs organizational average). Conversely, **Marketing** personnel demonstrated the highest average performance appraisal score (**4.17**), while generating the largest share of commercial sales (₹1.22M, 23.6%).

5. High Return on Employee Compensation (Salary ROI)

Cross-functional attribution demonstrates outstanding commercial leverage among select individual contributors. Employee Amit_82 (HR Executive, ₹40.9K salary) achieved an extraordinary **2.02x Salary ROI** (₹82.68K sales). Similarly, top billing producer Manav_59 (IT Data Analyst, ₹73.8K salary) led the firm with ₹111,364 in sales across 18 transactions (1.51x ROI).

10. Strategic Business Recommendations

1. **Replicate Western Margin Governance to High-Volume Territories:** The West region's margin efficiency (15.86%) should serve as the operational benchmark for the South region (14.82%). By enforcing stricter discount concession thresholds in the South, the firm can recapture an estimated ₹14,000+ in annual profit without sacrificing volume.
2. **Software & Services Solution Bundling:** With Software and Services delivering over 44% of total turnover at solid 15.3% margins, the product management team should design bundled cross-functional packages (Software Licenses + Implementation Services) to increase Average Order Value (AOV).
3. **Rebalance HR and Marketing Compensation Structures:** Address the structural compensation lag observed in the HR and Marketing departments. Marketing drives 23.6% of organizational revenue with an appraisal score of 4.17; implementing performance-linked quarterly sales bonuses will protect top talent from competitive attrition.
4. **Tiered Discount Governance:** Transactions with discount rates > 15% should require automated managerial sign-off. High discount rates currently erode margin on orders exceeding ₹8,000 without demonstrating measurable volume elasticity.
5. **Executive BI Self-Service Adoption:** Embed the Power BI dashboard into quarterly operational cadences. Utilizing the multi-attribute slicers (Year, City, Category, Department) allows regional directors to identify localized demand drops in real time.

11. Conclusion & Future Enhancement Roadmap

The **HR & Sales Performance Company Overview Analysis** project establishes an end-to-end analytics foundation connecting transactional commercial data with internal human capital metrics. By validating raw assets in Microsoft Excel, deploying 26 production queries in PostgreSQL / SQL, and engineering an interactive dashboard in Microsoft Power BI, this project equips decision-makers with a robust framework for evidence-based governance.

Future Scope: Upcoming iterations will integrate predictive customer churn algorithms via Python machine learning models and automate daily data pipeline refreshes via Microsoft Fabric / Power BI Gateway schedules.